

CTR Monitoring System

1. Development Duration

- **Total duration:** 47 days (extended work due to 8mm operational workload)
 - 40 half days
 - 7 full days

2. Purpose

To eliminate fake or unverifiable CTR reporting by introducing:

- Time-enforced CTR marking
- Automated activity verification
- Platform-specific validation logic
- Admin-level transparency

3. Technology Stack

Layer	Technology
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB
Authentication	JWT (JSON Web Tokens)
Automation	Selenium
Scheduling	Node-Cron
APIs	Reddit API, Apify (Instagram, Quora)
Tools	Postman, VS Code
Deployment	Github, Render
	BHW - Local (Current) / Cloud (Planned)

Database, Hosting, Github all connected with gvp.prakashparmar@gmail.com

4. System Architecture

The system follows a **two-step validation model**:
Employee intent + automated verification.

Architecture Flow

Employee → Marks CTR
↓
Task Execution stored
↓
Cron Job runs
↓
Platform API / Scraper
↓
Delta Calculation
↓
CTR Status Generated
↓
Admin Dashboard

Key Design Principle

- **Employee marking CTR does not mean completion**
- **Only cron verification decides final status**

5. Core Features

Employee Features

- Secure login with JWT authentication
- Daily CTR marking per account
- Platform-wise task visibility
- Account management
- Real-time status feedback after verification

Admin Features

- Centralized dashboard
- Platform-wise CTR monitoring
- Retry control and cooldown enforcement
- Cron-verified CTR status
- Partial and suspicious activity detection
- User and account management

6. Platform-Wise Working

Each platform has **custom verification logic**, based on what data is realistically available.

6.1 Instagram CTR

Verification Logic

- Baseline snapshot captured Follower/post delta calculated
- CTR must be marked before verification
- Time-based enforcement prevents post-CTR activity misuse

Possible Status

- Done
- Partial
- Not Done

CTR Monitor

Dashboard

Add Accounts

Instagram Accounts

Reddit Accounts

Quora Accounts

BHW Accounts

Daily Tasks

Keywords

Welcome, Prakash Parmar

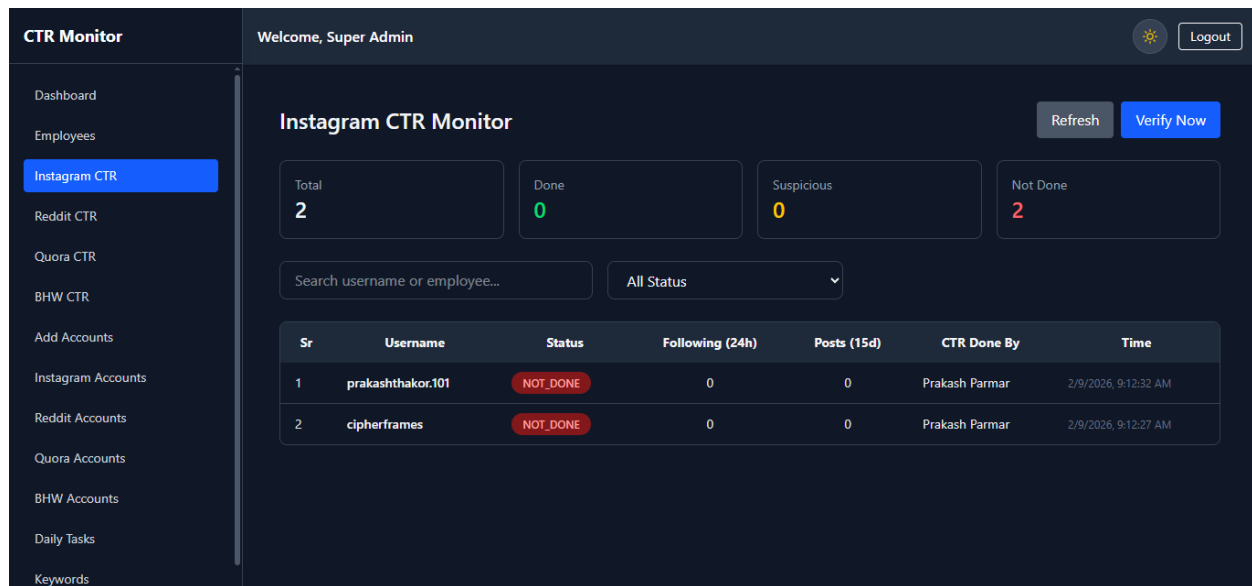
 Logout

Instagram Accounts

Total: 2 Working: 2 Suspicious: 0 Not Working: 0

Search by username...

Sr	Name	Username	Password	Link	Status	CTR Done
1	Prakash	cipherframes	demo-pass	https://www.instagram.com/cipherframes/	Working	☒
2	Prakash Parmar	prakashthakor.101	demo	https://www.instagram.com/prakashthakor.101/	Working	☒



6.2 Reddit CTR

Workflow

1. Employee marks CTR (intent only)
2. CTR record remains hidden from admin dashboard
3. Cron job runs at scheduled time
4. Reddit API verifies:
 - o Comments in last 24 hours
 - o Posts within last 15 days
5. Status generated after verification

Key Safeguard

- Admin **does not see CTR result until cron finishes**
- Prevents false “Not Done” visibility

Possible Status

- Done
- Suspicious
- Not Done

CTR Monitor

Welcome, Prakash Parmar

Logout

Dashboard
Add Accounts
Instagram Accounts
Reddit Accounts
Quora Accounts
BHW Accounts
Daily Tasks
Keywords

Reddit Accounts

Total: 2
Working: 2
Suspicious: 0
Not Working: 0

Search by User ID, Name or Email...

Sr	User ID	Name 1	Email 1	Pass 1	Name 2	Email 2	Pass 2	Status	CTR Done
1	101	Dramatic-Lobster-969	gvp.prakashparmar@gmail.com	demo	Dramatic-Lobster-969	gvp.prakashparmar@gmail.com	demo	Working	☒
2	12	ac_is_just_a_suggestion	RashadaCarpinelli@gmail.com	abc	poha_with_ketchup	RashadaCarpinelli@gmail.com	fe	Working	☒

CTR Monitor

Welcome, Super Admin

Logout

Dashboard
Employees
Instagram CTR
Reddit CTR
Quora CTR
BHW CTR
Add Accounts
Instagram Accounts
Reddit Accounts
Quora Accounts
BHW Accounts
Daily Tasks
Keywords

Reddit CTR Monitor

Refresh
Verify Now

Search username or employee...
All Status

Username	Status	Comments (24h)	Posts (15d)	CTR Done By	Time
ac_is_just_a_suggestion poha_with_ketchup	NOT DONE	0 / 4 0 + 0	0 / 1	Prakash Parmar	09 Feb, 09:19 am
Dramatic-Lobster-969 Dramatic-Lobster-969	SUSPICIOUS	0 / 4 0 + 0	1 / 1	Prakash Parmar	09 Feb, 09:19 am

6.3 Quora CTR

Verification Method

- Profile data fetched using Apify
- Metrics tracked:
 - Answers
 - Questions
 - Following
- Baseline comparison ensures correctness

Zero-Data Protection

- If Apify returns all zero values:
 - Data is ignored

- Database write is skipped
- Prevents false negatives

CTR Monitor

- Dashboard
- Add Accounts
- Instagram Accounts
- Reddit Accounts
- Quora Accounts**
- BHW Accounts
- Daily Tasks
- Keywords

Welcome, Prakash Parmar

Quora Accounts

Total: 1

Working: 1

Suspicious: 0

Not Working: 0

Search by User ID, Name or Email...

Sr	User ID	Name 1	Email 1	Pass 1	Name 2	Email 2	Pass 2	Status	CTR Done
1	101	Prakash-Parmar-826	prakashparmar1503@gmail.com	pass1	Prakash-Parmar-826	prakashparmar1503@gmail.com	pass2	Working	

CTR Monitor

- Dashboard
- Employees
- Instagram CTR
- Reddit CTR
- Quora CTR**
- BHW CTR
- Add Accounts
- Instagram Accounts
- Reddit Accounts
- Quora Accounts
- BHW Accounts
- Daily Tasks
- Keywords

Welcome, Super Admin

Quora CTR Monitor

Total

Done

Suspicious

Not Done

Pending

1

0

0

1

0

Search user ID or employee...

All Status

#	User ID	Status	Following (24h)	Answers (24h)	Questions (15d)	CTR Done By	Time
1	101	NOT_DONE	0	0 / 1	0	Prakash Parmar	2/9/2026, 9:23:45 AM

6.4 BlackHatWorld (BHW) CTR

Verification Method

- Selenium-based profile scraping
- Metrics tracked:
 - Message count delta
 - Thread activity (where possible)
 - Reaction score (display only)

Important Design Decision

- Reactions are **informational only**
- Reactions never decide CTR status

Retry Handling

- Cooldown enforced
- Worker health checked before retry

CTR Monitor

Dashboard

Add Accounts

Instagram Accounts

Reddit Accounts

Quora Accounts

BHW Accounts

Daily Tasks

Keywords

Welcome, Prakash Parmar

Logout

BHW Accounts

Total: 1

Working: 1

Suspicious: 0

Not Working: 0

Search by User ID, Name or Email...

Sr	User ID	Name	Email	Password	Link	Status	Actions
1	101	Prakash Parmar	prakashparmar1503@gmail.c...	demo i	https://www.blackhatworld.com/me...	Working	

CTR Monitor

Dashboard

Employees

Instagram CTR

Reddit CTR

Quora CTR

BHW CTR

Add Accounts

Instagram Accounts

Reddit Accounts

Quora Accounts

BHW Accounts

Daily Tasks

Keywords

Welcome, Super Admin

Logout

BHW CTR Monitor

Refresh

Total1

Done0

Partial0

Failed0

Pending0

Search username...

All Status

#	Username	Status	Messages	Reactions	CTR Done By	Time	Retries	Last Retry	Action
1	prakash07dz.2272143	NOT_DONE	0 / 1	2	Prakash Parmar	2/9/2026, 9:24:35 AM	1	2/9/2026, 9:27:47 AM	Retry

7. CTR Status Logic

Status	Meaning
Done	All required activity completed
Partial	Some activity detected
Suspicious	Activity detected but below expectation
Not Done	No valid activity
Pending	CTR marked but cron not yet run

Upcoming Project Costs

Database MongoDB (Atlas) – **Free tier**

Hosting – Render – **Free tier**

External Services

- **Apify**
 - Paid scraper usage for Instagram and Quora
 - Cost scales with run frequency

Infrastructure

- **BHW Selenium Worker**
 - VPS required for stable scraping
 - Dedicated server recommended for scale

These costs are **planned for future deployment**, not required during initial development.

Future Enhancements

Error Handling & Schema Validation

- Centralized error handling
- Strict MongoDB schema validation
- Prevents invalid writes and silent failures

Why Necessary

- Improves system stability
- Easier debugging and maintenance

Database Optimization

- Automatic deletion of old CTR records
- Retain only:
 - Recent daily records
 - Last 15 days for post validation

Why Necessary

- Reduces database size
- Controls cloud hosting costs
- Improves query performance

Testing (Optional)

Unit Testing

- **Jest**
- Validate:
 - Controllers
 - Status calculation logic
 - Utility functions

End-to-End Testing

- **Cypress**
- Validate:
 - Login flow
 - CTR marking
 - Admin dashboard behavior

Live URL - <https://gvp-ctrmanager.onrender.com>

Admin Credentials (Temp) – username - admin

Password – admin123